Area of a circle

1

a
- i $64\pi \text{ cm}^2$
- ii 201.06 cm^2

b
- i $49\pi \text{ cm}^2$
- ii 153.94 cm^2

c
- i $25\pi \text{ cm}^2$
- ii 78.54 cm^2

d
- i $\frac{169\pi}{4} \text{ cm}^2$
- ii 132.73 cm^2

e
- i $324\pi \text{ cm}^2$
- ii 1017.88 cm^2

f
- i $16\pi \text{ cm}^2$
- ii 50.27 cm^2

2

a $16\pi \text{ mm}^2$ **b** $16\pi \text{ mm}^2$ **c** $144\pi \text{ cm}^2$

3

a 254.47 cm^2 **b** 452.39 cm^2 **c** 452.39 cm^2

4

a
- i 6 mm
- ii $36\pi \text{ mm}^2$

b
- i 7 cm
- ii $49\pi \text{ cm}^2$

c
- i $\frac{9}{\pi} \text{ cm}$
- ii $\frac{81}{\pi} \text{ cm}^2$

d
- i 11 mm
- ii $121\pi \text{ cm}^2$

5

a $r = 8 \text{ cm}$ **b** $r = \frac{6}{\sqrt{\pi}} \text{ cm}$ **c** $r = 6 \text{ cm}$ **d** $r = 4 \text{ cm}$

e $r = \frac{5}{\sqrt{\pi}} \text{ mm}$

6

a i 5 cm ii $10\pi \text{ cm}$

b i 11 cm ii $22\pi \text{ cm}$

7

a $r = 10.59$ **b** 66.5 cm

8

a $\frac{12}{\sqrt{\pi}} \text{ mm}$ **b** 6.77 mm

9

a 4 cm **b** 8 cm

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10

a $\frac{5}{\sqrt{\pi}}\,\text{mm}$ **b** $\frac{10}{\sqrt{\pi}}\,\text{mm}$

11

a $\frac{24}{\sqrt{\pi}}\,\text{mm}$ **b** $13.54\,\text{mm}$

12

a $14.1\ \text{cm}^2$ **b** $19.6\ \text{mm}^2$ **c** $95.0\ \text{m}^2$ **d** $115.5\ \text{cm}^2$

Applications

13 $314.16\ \text{cm}^2$

14

a $40\ \text{m}$ **b** $251.33\ \text{m}$ **c** $5026.55\ \text{m}^2$

15

a $399\,832.26\ \text{cm}^2$

b $399\,629.57\ \text{cm}^2$

c Yes, because the approximate area is $202.69\ \text{cm}^2$ smaller than the actual area.

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